

Goal

Preparation of an NMR sample under argon atmosphere. The sample is always mixed with the deuterated NMR solvent in a separate Schlenk to facilitate the dilution of the sample and the formation of a homogeneous solution. If not mentioned otherwise always J. Young NMR tubes are used to obtain the highest security for the sample.

Prerequisites and preparation

J Young tube (also possible with NMR tubes with screw caps)

"Schlenk line - J Young tube transition piece"

flame dried empty Schlenk

Steps

Step number	Step description
1. Fill sample in Schlenk	The sample (ca 20.mg) is filled into a flame dried Schlenk tube. Version a) The sample is filled at air into a the tube, if it is stable enough at air. after the transfer the schlenk tube (and the sample soure) has to be placed under argon. Version b) Eventually the sample is stored inside a glovebox. Then a flame dried Schlenk tube, spectacular/pipette & pipette hat are infiltrated inside a glovebox in order to fill the sample in the Schlenk tube. After transferration of the sample into the tube inside the glovebox the tube is removed from the box and connected to the schlenk line
2. Adding NMR solvent	Degassed and dried deuterated NMR solvent ca. 0.75 mL is added into the Schlenk tube through a septum. (Septum on both schlenk tubes)
3. Prepare J Young tube	A J Young Tube is equipped with a transition piece to the Schlenk line. The transition pieces is equipped with a septum. The J. Young is carefully flame dried. Caution: Only the tube not the transition piece should be heated
4. Set J Young tube under vacuum	The flame dried J Young NMR Tube is set under vacuum and closed. The transition piece is open to argon while the closed NMR tube stayed under vacuum.
5. Transfer the solution into the transition piece	The sample solution is transferred via a flushed syringe into the transition piece of the J Young tube
6. Open the J Young tube.	The J Young tube is opened fast allowing the liquid to flow into the J Young tube. The Argon is still on to push the liquid inside the evacuated Young Tube.
7. Close the J Young tube	The J Young tube is closed and the transition piece as well as the cap of the Young tube is washed quickly

NMR-Measurements

¹H: ¹H or ¹H (12 ppm to - 30 ppm)

³¹P: ³¹P Hdec quant cryo with 64 scan

Comment

On 2022-06-30 09:39:54 Kristína Rabatinová wrote:

Add info when we leave the glovebox

0.6 mL → 0.7 mL



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